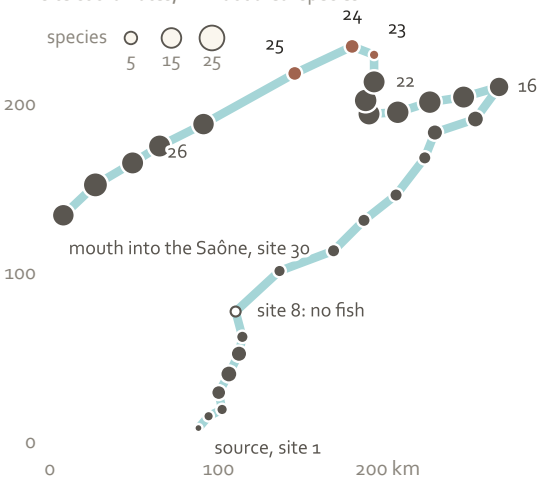
Doubs fish fall from 22 species to 3 at a polluted stretch, and recover below it

30 sites over 453 km, 27 species · terracotta: sites 23–25, ammonium up to 1.8 mg/L, oxygen down to 4.1 mg/L

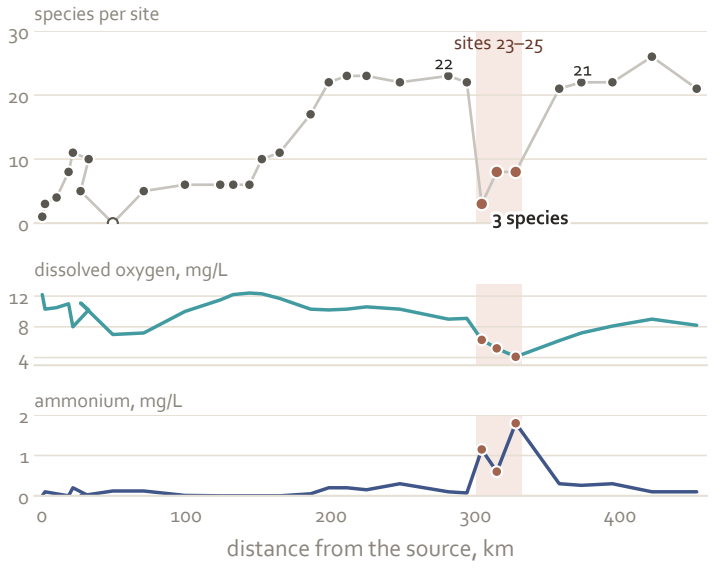
a The river, source to mouth

site coordinates, km · dot area: species



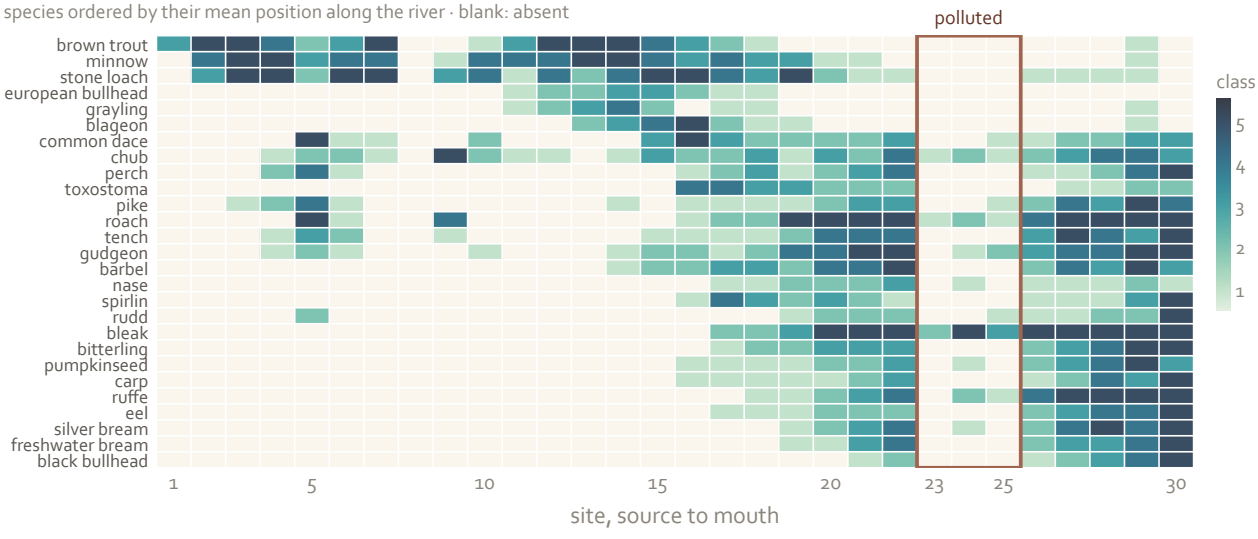
b Where oxygen drops and ammonium peaks, the fish go

species per site



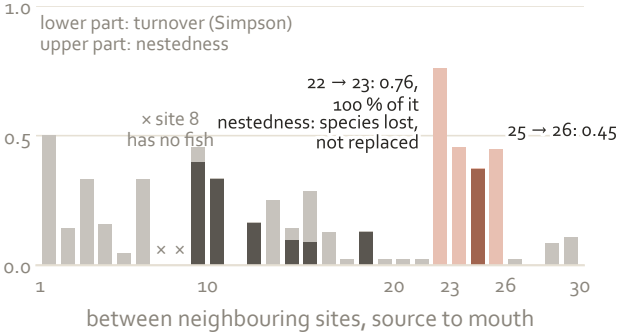
c Trout give way to cyprinids downstream; at 23–25 almost every species is missing

species ordered by their mean position along the river · blank: absent



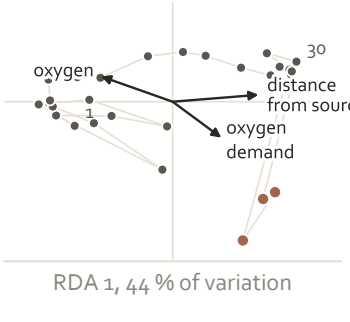
d Pollution removes species, it does not swap them

Sørensen dissimilarity to the next site



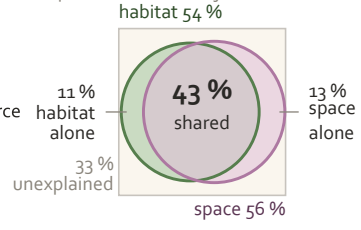
e Three variables explain 54 %

RDA 2, 10 % · sites joined in river order



f Space and habitat overlap

square: all · areas: adjusted R^2



Data: Doubs river fish (Verneaux 1973), via R ade4::doubs; analyses after Borcard, Gillet and Legendre (2018) · doubs_fish.py · git d3bf290